

2. (a) $u_1 = 12$

A1

[1 mark]

(b) $15 - 3n = -33$

(A1)

$n = 16$

A1

[2 marks]

(c) valid approach to find d

(M1)

$u_2 - u_1 = 9 - 12$ OR recognize gradient is -3 OR attempts to solve

$-33 = 12 + 15d$

$d = -3$

A1

[2 marks]

Total [5 marks]

Section A

1. (a) 1.01206..., 2.45230...

$$a = 1.01, b = 2.45 \text{ (} 1.01x + 2.45 \text{)}$$

A1A1

[2 marks]

- (b) 0.981464...

$$r = 0.981$$

A1

Note: A common error is to enter the data incorrectly into the GDC, and obtain the answers $a = 1.01700...$, $b = 2.09814...$ and $r = 0.980888...$. Some candidates may write the 3 sf answers, ie. $a = 1.02$, $b = 2.10$ and $r = 0.981$ or 2 sf answers, ie. $a = 1.0$, $b = 2.1$ and $r = 0.98$. In these cases award **A0A0** for part (a) and **A0** for part (b). Even though some values round to an accepted answer, they come from incorrect working.

[1 mark]

- (c) correct substitution of 78 into **their** regression equation

(M1)

$$81.3930... \text{ } 81.23 \text{ from 3 sf answer}$$

$$81$$

A1

[2 marks]

Total [5 marks]

3. METHOD 1 (finding u_1 first, from S_8)

$$4(u_1 + 8) = 8 \quad (\text{A1})$$

$$u_1 = -6 \quad \text{A1}$$

$$u_1 + 7d = 8 \text{ OR } 4(2u_1 + 7d) = 8 \text{ (may be seen with their value of } u_1) \quad (\text{A1})$$

attempt to substitute their u_1 (M1)

$$d = 2 \quad \text{A1}$$

METHOD 2 (solving simultaneously)

$$u_1 + 7d = 8 \quad (\text{A1})$$

$$4(u_1 + 8) = 8 \text{ OR } 4(2u_1 + 7d) = 8 \text{ OR } u_1 = -3d \quad (\text{A1})$$

attempt to solve linear or simultaneous equations (M1)

$$u_1 = -6, d = 2 \quad \text{A1A1}$$

[5 marks]

4. substitutes into $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ to form

$$0.55 = 0.4 + P(B) - P(A \cap B) \quad (\text{or equivalent}) \quad (\mathbf{A1})$$

$$\text{substitutes into } P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ to form } 0.25 = \frac{P(A \cap B)}{P(B)} \quad (\text{or equivalent}) \quad (\mathbf{A1})$$

attempts to combine their two probability equations to form an equation in $P(B)$ ($\mathbf{M1}$)

Note: The above two **A** marks are awarded independently.

correct equation in $P(B)$ ($\mathbf{A1}$)

$$0.55 = 0.4 + P(B) - 0.25P(B) \quad \text{OR} \quad \frac{P(B) - 0.15}{P(B)} = 0.25 \quad \text{OR} \quad P(B) - 0.15 = 0.25P(B)$$

(or equivalent)

$$P(B) = \frac{15}{75} \left(= \frac{1}{5} = 0.2 \right) \quad (\mathbf{A1})$$

Total [5 marks]

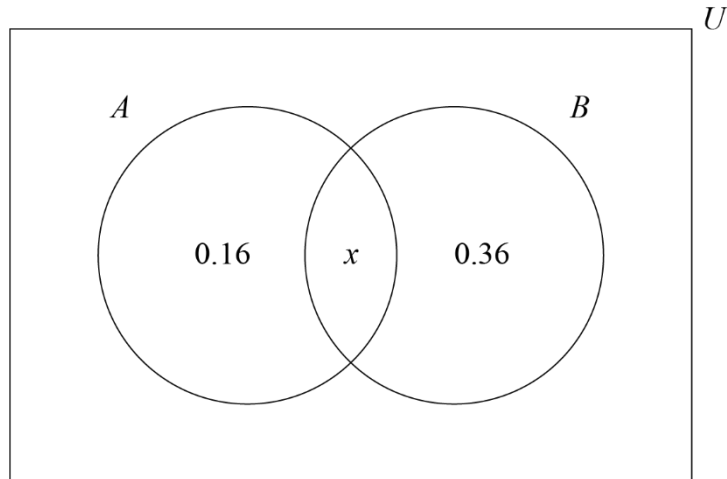
6. (a) **METHOD 1**

EITHER

one of $P(A) = x + 0.16$ OR $P(B) = x + 0.36$

A1

OR



A1

THEN

attempt to equate their $P(A \cap B)$ with their expression for $P(A) \times P(B)$

M1

$$P(A \cap B) = P(A) \times P(B) \Rightarrow x = (x + 0.16)(x + 0.36)$$

A1

$$x = 0.24$$

A1

METHOD 2

attempt to form at least one equation in $P(A)$ and $P(B)$ using independence

M1

$$(P(A \cap B') = P(A) \times P(B') \Rightarrow) P(A) \times (1 - P(B)) = 0.16 \text{ OR}$$

$$(P(A' \cap B) = P(A') \times P(B) \Rightarrow) (1 - P(A)) \times P(B) = 0.36$$

$$P(A) = 0.4 \text{ AND } P(B) = 0.6$$

A1

$$P(A \cap B) = P(A) \times P(B) = 0.4 \times 0.6$$

(A1)

$$x = 0.24$$

A1

[4 marks]

continued...

Question 6 continued

(b) **METHOD 1**

recognising $P(A' | B') = P(A')$

(M1)

$$= 1 - 0.16 - 0.24$$

$$= 0.6$$

A1

METHOD 2

$$P(B) = 0.36 + 0.24 (= 0.6)$$

$$P(A' | B') = \frac{P(A' \cap B')}{P(B')} \left(= \frac{0.24}{0.4} \right)$$

(A1)

$$= 0.6$$

A1

[2 marks]

Total [6 marks]

3. (a) valid approach to find area of segment by finding area of sector – area of triangle (M1)

$$\frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta$$

$$\frac{1}{2}(2)^2\theta - \frac{1}{2}(2)^2\sin\theta \quad (A1)$$

$$\text{area} = 2\theta - 2\sin\theta \quad A1$$

[3 marks]

- (b) EITHER

$$\text{area of logo} = \text{area of rectangle} - \text{area of segments} \quad (M1)$$

$$5 \times 4 - 2 \times (2\theta - 2\sin\theta) = 13.4 \quad (A1)$$

OR

$$\text{area of one segment} = \frac{20 - 13.4}{2} (= 3.3) \quad (M1)$$

$$2\theta - 2\sin\theta = 3.3 \quad (A1)$$

THEN

$$\theta = 2.35672\dots$$

$$\theta = 2.36 \text{ (do not accept an answer in degrees)} \quad A1$$

Note: Award (M1)(A1)A0 if there is more than one solution.

Award (M1)(A1FT)A0 if the candidate works in degrees and obtains a final answer of 135.030...

[3 marks]

Total [6 marks]

6. (a) **EITHER**

recognises the required term (or coefficient) in the expansion

(M1)

$$bx^5 = {}^7C_2 x^5 1^2 \quad \text{OR} \quad b = {}^7C_2 \quad \text{OR} \quad {}^7C_5$$

$$b = \frac{7!}{2!5!} \left(= \frac{7!}{2!(7-2)!} \right)$$

correct working

A1

$$\frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 5 \times 4 \times 3 \times 2 \times 1} \quad \text{OR} \quad \frac{7 \times 6}{2!} \quad \text{OR} \quad \frac{42}{2}$$

OR

lists terms from row 7 of Pascal's triangle

(M1)

1, 7, 21, ...

A1

THEN

$$b = 21$$

AG

[2 marks]

(b) $a = 7$

(A1)

correct equation

A1

$$21x^5 = \frac{ax^6 + 35x^4}{2} \quad \text{OR} \quad 21x^5 = \frac{7x^6 + 35x^4}{2}$$

correct quadratic equation

A1

$$7x^2 - 42x + 35 = 0 \quad \text{OR} \quad x^2 - 6x + 5 = 0 \quad (\text{or equivalent})$$

valid attempt to solve **their** quadratic

(M1)

$$(x-1)(x-5) = 0 \quad \text{OR} \quad x = \frac{6 \pm \sqrt{(-6)^2 - 4(1)(5)}}{2(1)}$$

$$x = 1, x = 5$$

A1

Note: Award final **A0** for obtaining $x = 0, x = 1, x = 5$.

[5 marks]

Total [7 marks]

5. (a) recognising $v = 0$ (M1)

$$t = 6.74416\dots$$

$$= 6.74 \text{ (sec)}$$

A1

Note: Do not award **A1** if additional values are given.

[2 marks]

(b) $\int_0^{10} |v(t)| dt$ OR $-\int_0^{6.74416\dots} v(t) dt + \int_{6.74416\dots}^{9.08837\dots} v(t) dt - \int_{9.08837\dots}^{10} v(t) dt$ (A1)

$$= 37.0968\dots$$

$$= 37.1 \text{ (m)}$$

A1

[2 marks]

- (c) recognizing acceleration at $t = 7$ is given by $v'(7)$ (M1)

$$\text{acceleration} = 5.93430\dots$$

$$= 5.93 \text{ (ms}^{-2}\text{)}$$

A1

[2 marks]

Total [6 marks]

6. (a) translation (shift) by $\frac{3\pi}{2}$ to the right/positive horizontal direction **A1**
 translation (shift) by q upwards/positive vertical direction **A1**

Note: accept translation by $\begin{pmatrix} \frac{3\pi}{2} \\ q \end{pmatrix}$

Do not accept 'move' for translation/shift.

[2 marks]

(b) **METHOD 1**

minimum of $4\sin\left(x - \frac{3\pi}{2}\right)$ is -4 (may be seen in sketch) **(M1)**

$$-4 + 2.5 + q \geq 7$$

$q \geq 8.5$ (accept $q = 8.5$) **A1**

substituting $x = 0$ and their q ($= 8.5$) to find r **(M1)**

$$(r =) 4\sin\left(\frac{-3\pi}{2}\right) + 2.5 + 8.5$$

$$4 + 2.5 + 8.5$$
 (A1)

smallest value of r is 15 **A1**

continued...

Question 6 continued

METHOD 2

substituting $x = 0$ to find an expression (for r) in terms of q **(M1)**

$$(g(0) = r =) 4\sin\left(\frac{-3\pi}{2}\right) + 2.5 + q$$

$$(r =) 6.5 + q \quad \text{A1}$$

minimum of $4\sin\left(x - \frac{3\pi}{2}\right)$ is -4 **(M1)**

$$-4 + 2.5 + q \geq 7$$

$$-4 + 2.5 + (r - 6.5) \geq 7 \quad (\text{accept } =) \quad \text{A1}$$

smallest value of r is 15 **A1**

METHOD 3

$$4\sin\left(x - \frac{3\pi}{2}\right) + 2.5 + q = 4\cos x + 2.5 + q \quad \text{A1}$$

y -intercept of $4\cos x + 2.5 + q$ is a maximum **(M1)**

amplitude of $g(x)$ is 4 **(A1)**

attempt to find least maximum **(M1)**

$$r = 2 \times 4 + 7$$

smallest value of r is 15 **A1**

[5 marks]

Total [7 marks]